



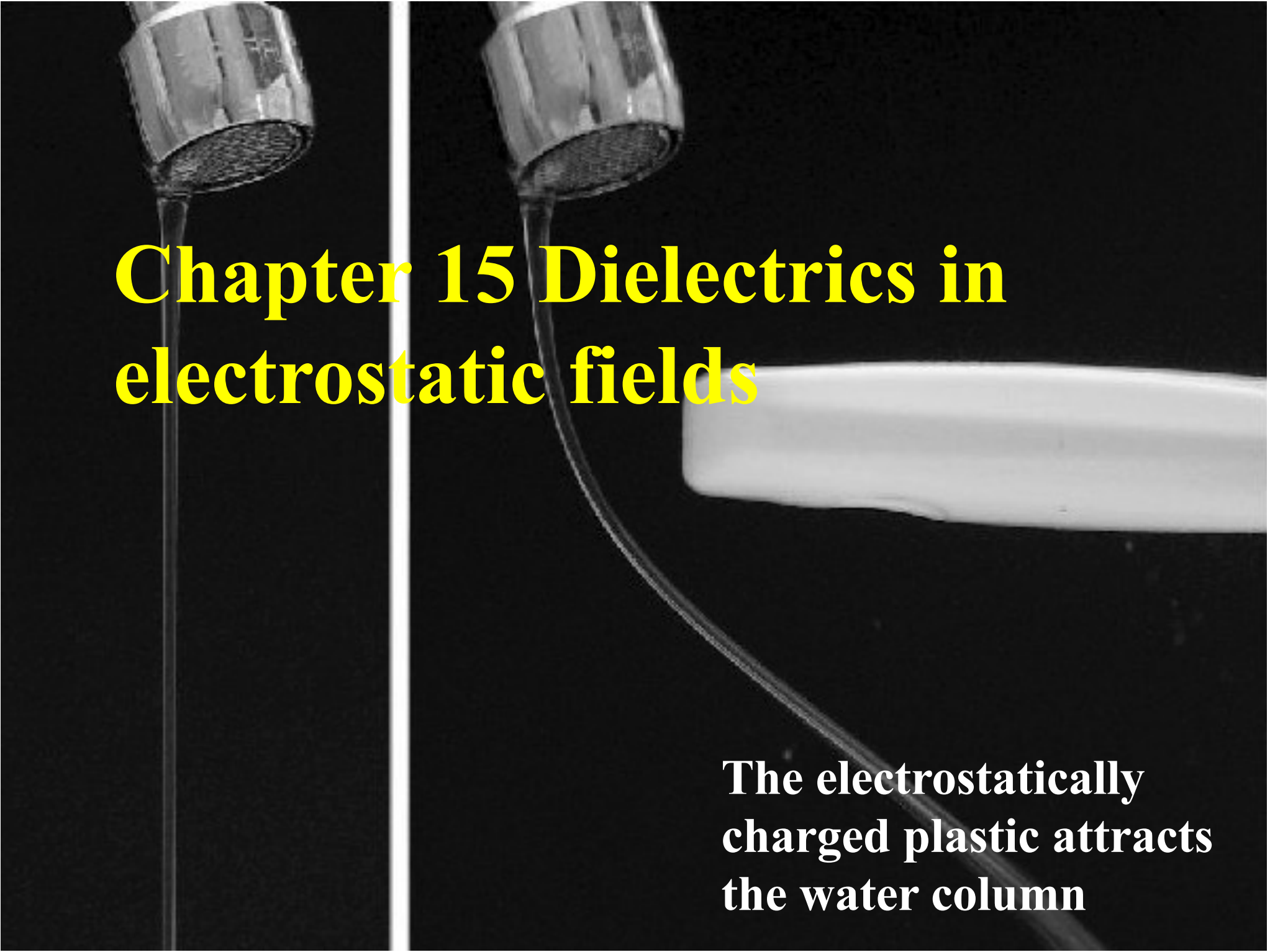
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Chapter 15 Dielectrics in electrostatic fields

The electrostatically charged plastic attracts the water column

Chapter 15 Dielectrics in electrostatic fields

§ 15.1 Effect of dielectric on electric field

§ 15.2 Polarization of dielectrics

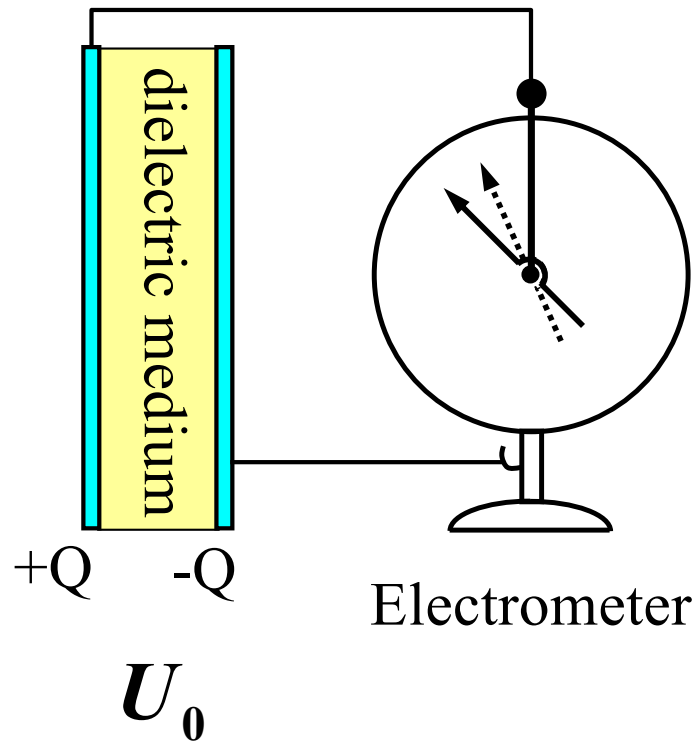
§ 15.3 The laws of electrostatic field in the presence of medium

§ 15.4 Capacitors and their capacitance

§ 15.5 Electrostatic field energy in the presence of a medium

§ 15.1 Effect of dielectric on electric field

Polarization of dielectrics under electric fields and their influence on electric fields



When the dielectric medium is inserted, the angle of the electrometer pointer decreases, indicating that the electric field between the plates is weakened.

$$U = U_0 / \epsilon_r \quad E = E_0 / \epsilon_r$$

ϵ_r : relative dielectric constant

Dielectric constants of common dielectrics

Material	Temperature (Celsius)	Relative dielectric constant
vacuum	20	1
water	14	81.5
water vapor	140-150	1.00785
air	20	1.000585
glycerin		45.8
kerosene	20	2-4
paper		5
glass		5~10
barium dioxide		106
barium titanate		$10^3 \sim 10^4$

Effect of dielectric on electric field

Dielectric: Unlike a conductor, it is an electrical insulator that has no free charge, no conduction.

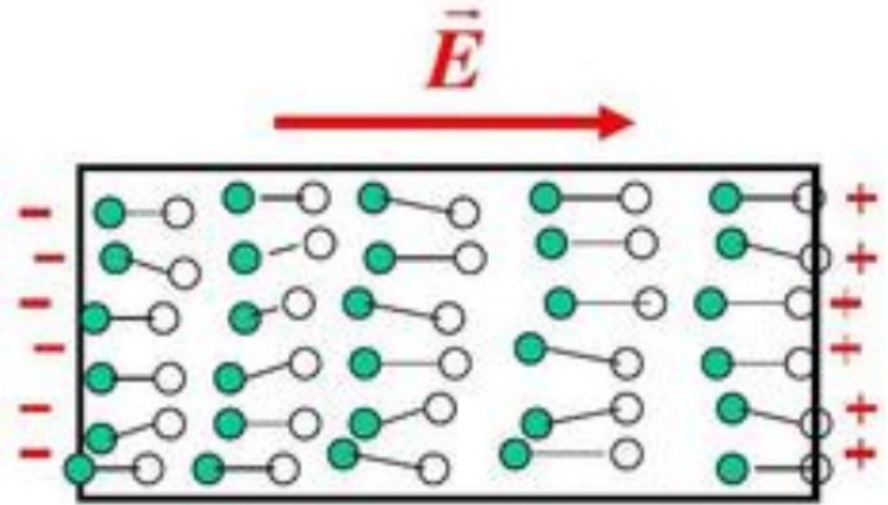
If a dielectric is placed in an electrostatic field, the electrons and nuclei in the dielectric atoms will make microscopic relative displacements within the atomic range under the action of the electric field force.

When electrostatic equilibrium is reached, the field strength inside the dielectric is not zero.

In an external electric field, the dielectric is affected by the electric field and also affects the external electric field.

§ 15.2 Polarization of dielectrics

The phenomenon that an insulating dielectric exhibits an electrical effect under the action of an electric field is called polarization of the dielectric.



Polarized charges that cannot move freely appear in polarized dielectrics or at their interfaces, and are also called bound charges.

Unlike free charges in conductors, the electric field generated by polarized charges in a dielectric cannot completely offset the external electric field.

Why?

Microscopic electrical structure of dielectrics

Atoms and molecules are composed of atomic nuclei and electron clouds. When considering the electric field they generate at a distance, as a simplification, all positive (negative) charges can be equivalent to a point charge with positive (negative) charge, and its position is called the positive (negative) charge center.



$$\text{Molecular dipole moment } p = ql$$